Guo Ye

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Summary

Machine learning researcher building **AI-driven protein engineering workflows** end to end: generative and structure-prediction models for protein and ligand design, the data engineering and GPU infrastructure that makes them reproducible at scale, and the laboratory automation that closes the design–build–test–learn loop with a wet lab. I build generative-model pipelines for binder design plus a robotic front end for automated synthesize. My PhD work in deep learning gives me years of grounding in **closed-loop autonomous systems** — perception, decision, cloud/edge deployment, monitoring and recovery.

Education

2020 – 2026 Ph.D. (Computer Science), **Northwestern University**. Advised by Prof. Han Liu

2018 – 2020 M.S. Computer Science, McCormick School of Engineering, **Northwestern University** (3.78/4.0)

2014 – 2018 B.S. Computer Science, Yingcai Honors School, **University of Electronic Science and Technology of China** (3.87/4.0)

Technical Skills

Protein / molecular design AF3-style co-folding (Protenix, Boltz, PXDesign, OpenDDE); inverse folding (ProteinMPNN / LigandMPNN); protein language models (ESM, PLMSearch); latent-diffusion generative design; docking, multi-state filtering, hotspot analysis

Machine learning PyTorch; diffusion models, VAEs, transformers; Reinforcement Learning, imitation learning and active learning; foundation-model fine-tuning, transfer and catastrophic-forgetting control; benchmark design and OOD evaluation

Data engineering Dataset curation, deduplication and provenance across heterogeneous public corpora; experiment tracking (MLflow, Kubeflow); structure and assay data pipelines

Scientific computing Multi-GPU training (A100 / H100 / RTX 6000 Ada); AWS, Kubernetes, Docker, Ray, SLURM; CI/CD for model deployment

Agentic AI & interfaces LLM agent pipelines, automated analysis-to-report and interactive-visualization generation; Python, C/C++, Java, MATLAB, R

Lab automation Wet lab workflows; Hamilton liquid-handling integration; dual-arm manipulation, LeRobot, ROS; simulation (IsaacGym, PyBullet, Gazebo)

AI for Protein Engineering

Generative Models for Targeted Protein 2026 – present

- Design and build an end-to-end computational discovery pipeline: generative candidate proposal, physics- and chemistry-based filtering, and a scoring model fit on wet lab data.
- Fine-tuned a block-level latent-diffusion model for unified small-molecule / peptide / binder design on public corpora, using sequential VAE-then-diffusion pipeline to settle a disputed design decision.

Co-folding & Binder-Design Model Infrastructure 2026 – present

- Deploying a suite of AF3-class models (Protenix, Boltz, PXDesign, OpenDDE) as interchangeable components behind one design interface.
- Built a full de novo protein design pipeline that chains RFdiffusion, LigandMPNN, ESM, AutoDock and XGBoost into one automated design–build–test–learn loop.
- Built a protein-language-model embedding search (PLMSearch) for binding proteins, recovering homologs that sequence-identity search misses and delivering ranked, annotated results.

Automated Wet Lab 2025 – 2026

- Automating protein expression with a dual-arm `AgileX` manipulator trained by imitation learning (LeRobot), covering dexterous steps that fixed-function liquid handlers cannot reach.
- Designed a DNA preparation workflow as the robotic front end feeding a Opentron liquid-handling system in the simulator so sample prep and expression run as one automated, logged pipeline.
- Instrumented the workflow to log every action, image and instrument reading to a structured store keyed by run and sample, so wet-lab output arrives as model-ready training data rather than spreadsheets to be reconciled after the fact.

Autonomous Systems & ML Infrastructure

DOS®: A Deployment Operating System for Autonomous Agents Jan 2022 – Sep 2022

– Built DOS®, a cloud/edge MLOps platform on AWS for reliably operating data-driven autonomous agents in production and simulation: CI/CD for model rollout, versioned storage reservoirs, failure recovery, and packaged executables — the same warehousing-and-orchestration layer an automated cloud lab needs between its instruments and its models.

– Adopted by lab members to move from ad-hoc scripts to reproducible, fully automated operation; demonstrated on a swarm of mobile robots. Paper under review.

Composing Pre-trained Foundation Models for Closed-Loop Control Dec 2023 – Sep 2024

– Developed an end-to-end algorithm that treats the outputs of several large frozen perception models as "prompts" into a decision transformer — a general recipe for reusing pre-trained models across a suite of downstream tasks without retraining each one.

– Deployed on a `Unitree GO1` across meadow, mud and ladder terrain, demonstrating simulation-to-reality transfer of the learned policy.

Search & Multi-Agent Learning in High-Dimensional Spaces 2020 – 2022

– Proposed a meta path-planning algorithm that transfers experience across problems in high-dimensional continuous state and action spaces, cutting the cost of each new search instance (*ICLR 2021*); and a path-following adversarial training strategy for robust decentralized collision avoidance under a Markov game, validated on hardware (*ICRA*).

Publications

Guo Ye, Muye Jia, Mingfu Liang, Matthew Elwin, and Han Liu (2024). "Visual Prompt via Pre-trained Model for Quadrupedal Locomotion". In: Submmitted to ICRA2025.

Guo Ye, Qinjie Lin, Zening Luo, and Han Liu (2024). "DOS®: A Deployment Operating System for Robots". In: 2024 IEEE International Conference on Robotics and Automation (ICRA).

Lin, Qinjie, Guo Ye, and Han Liu (2023). "EMS®: A Massive Computational Experiment Management System towards Data-driven Robotics". In: *2023 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, pp. 11338–11344.

Luo, Haozheng, Ruiyang Qin, Chenwei Xu, Guo Ye, and Zening Luo (2023). "Open-Ended Multi-Modal Relational Reasoning for Video Question Answering". In: *2023 32nd IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*. IEEE, pp. 363–369.

Guo Ye, Qinjie Lin, Wanxin Jin, Zhaoran Wang, Haozheng Luo, Tim Tsz-Kit Lau, cheng zhou, and Han Liu (2022). "RobLAX: A Differentiable Robotics Framework for Physics Augmented Reinforcement Learning". In: Submmitted. URL: https://sites.google.com/u.northwestern.edu/roblax/home.

Guo Ye, Han Liu, and Biswa Sengupta (2022). "Learning to Infer Belief Embedded Communication". In: *arXiv preprint arXiv:2203.07832*.

Lin, Qinjie, Guo Ye, Jiayi Wang, and Han Liu (2022). "RoboFlow: a Data-centric Workflow Management System for Developing AI-enhanced Robots". In: *Conference on Robot Learning*. PMLR, pp. 1789–1794.

Guo Ye, Qinjie Lin, Tzung-Han Juang, and Han Liu (2020). "Collision-free Navigation of Human-centered Robots via Markov Games". In: *2020 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, pp. 11338–11344.

Chen, Binghong, Bo Dai, Qinjie Lin, Guo Ye, Han Liu, and Le Song (2020). "Learning to plan in high dimensions via neural exploration-exploitation trees". In: *2020 International Conference on Learning Representations (ICLR)*.

He, Shiyuan, Guo Ye, Mengqiu Hu, Yang Yang, Fumin Shen, Heng Tao Shen, and Xuelong Li (2018). "Learning binary codes with local and inner data structure". In: *Neurocomputing* 282, pp. 32–41.